

Module 1: Introduction & Energy

1.1 Important Definitions

1. Define: Prime Mover, System, Boundary, Surroundings. (W-19, W-20, S-24, W-24)
2. Define: State, Path, Process, Cycle. (W-20, W-23)
3. State Zeroth Law & First Law of Thermodynamics (Mathematical Form). (W-21, W-20, W-24, S-25)
4. Define Specific Heat (C_p , C_v), Enthalpy, Internal Energy. (W-22, S-22, S-25)

1.2 Theory Questions

1. **Thermodynamic Systems:** Explain Open, Closed, and Isolated systems with neat sketches. Give examples. (W-21, S-20, W-18, W-24, S-24, S-25)
2. **Energy Sources:** Differentiate between Renewable and Non-Renewable energy sources. List examples. (W-22, S-24, S-25, W-24) [Image of Renewable Energy Sources]
3. **Global Warming:** Explain Global Warming and Ozone Depletion. List primary causes and impacts. (W-21, S-23, W-20, W-24, S-25)
4. **Gas Constant Relation:** With usual notations, prove that $C_p - C_v = R$. (W-21, S-22, S-25)
5. **First Law:** Write the mathematical statement of the First Law for a stationary closed system and explain the terms (Heat vs Work). (S-25, W-23)

Module 2: Properties of Gases & Steam

2.1 Important Definitions

1. Define: Boyles Law, Charless Law, Gay-Lussac's Law, Avogadros Law. (W-20, W-24)
2. Define: Dryness Fraction, Wet Steam, Superheated Steam, Critical Point. (W-21, W-22, S-24, W-24, S-25)
3. Define: Sensible Heat, Latent Heat, Degree of Superheat, Enthalpy of Evaporation. (S-22, S-24, W-24)

2.2 Theory Questions

1. **Steam Formation:** Describe steam formation from water at 0°C to superheated steam using a $T - h$ diagram. (W-21, W-18, W-24)
2. **Processes:** Explain Isothermal, Adiabatic, and Polytropic processes. Derive expressions for Work Done & Internal Energy. (W-21, S-22, W-20, S-24, W-24, S-25)
3. **Calorimeters:** Explain construction and working of:
 - **Throttling Calorimeter** (W-22, S-25, W-24)

- **Separating Calorimeter** (S-24, W-20)
- **Barrel Calorimeter** (W-23)

4. **Gas Equation:** Derive the Characteristic Gas Equation $PV = mRT$. (S-22, W-24, S-25)

2.3 Typical Numericals

1. **Gas Laws:** Gas initially at specific P, V, T undergoes compression/expansion. Find Final Temp, Work Done, Change in Enthalpy/Internal Energy. (S-23, S-24, W-24, W-23, S-25)
2. **Steam Properties:** Calculate enthalpy, volume, entropy, and internal energy for wet/superheated steam given pressure and dryness fraction. (S-23, W-19, W-24, S-25)
3. **Calorimeter:** Determine dryness fraction using Separating & Throttling calorimeter data. (W-24)

Module 3: Heat Engines & IC Engines

3.1 Important Definitions

1. Define: Heat Engine, Thermal Eff., Mechanical Eff. (S-23, S-25)
2. Define: Swept Volume, Clearance Volume, Compression Ratio (TDC/BDC). (W-20, S-24)
3. Define: Indicated Power, Brake Power, Specific Fuel Consumption. (S-23, S-24)

3.2 Theory Questions

1. **Otto Cycle:** Derive Air Standard Efficiency of Otto Cycle. (S-20, W-22, W-19, W-23, S-24, W-24, S-25) [Image of PV diagram of Otto cycle]
2. **Diesel Cycle:** Derive Air Standard Efficiency of Diesel Cycle. (S-25, W-24, S-24)
3. **Comparisons:**
 - Two-Stroke vs. Four-Stroke Engines. (W-21, W-24)
 - Petrol (SI) vs. Diesel (CI) Engines. (W-21, W-24, S-25)
4. **Components:** Function of Spark Plug, Injector, Piston Rings, Carburetor. (W-23, S-24, S-25)
5. **Rankine Cycle:** Derive thermal efficiency and draw T-s diagram. (W-18, W-23, W-24)
6. **Carnot Cycle:** State assumptions and limitations. (W-24)

3.3 Typical Numericals

1. **Cycle Efficiency:** Calculate efficiency for Otto/Diesel/Carnot cycles given temperatures or compression ratios. (W-22, S-24, S-25)
2. **IC Engine Performance:** Calculate BP, IP, Brake Thermal Efficiency, Fuel Consumption from trial data (Morse test/Rope brake). (Recurring: W-21, W-22, S-23, W-23, S-24, W-24, S-25)

Module 4: Steam Boilers

4.1 Theory Questions

1. **Classification:** Fire Tube vs. Water Tube boilers. (W-22, S-23, W-19, S-24, S-25)
2. **Specific Boilers:** Explain construction and working with neat sketch:
 - **Cochran Boiler** (W-22, W-19, W-24, S-25)
 - **Babcock & Wilcox Boiler** (W-21, S-22, W-24, S-25)
 - **Lancashire Boiler** (S-24)
3. **Mountings & Accessories:**
 - Differentiate between Mountings and Accessories. (W-24, W-23)
 - Explain: Water Level Indicator, Fusible Plug, Safety Valve, Pressure Gauge. (S-20, W-24)
 - Explain: Economizer, Air Preheater, Superheater. (S-20, W-23, S-24)
4. **Requirements:** List essential qualities of a good boiler. (W-24, S-25)

Module 5: Compressors & Pumps

5.1 Theory Questions

1. **Compressors:**
 - Differentiate Reciprocating vs. Rotary Compressors. (S-20, S-25)
 - Explain Centrifugal Compressor / Roots Blower. (S-24)
 - Need for Multistaging in air compressors. (W-18, W-24)
 - Definitions: FAD, Swept Volume, Volumetric Efficiency. (W-24)
2. **Pumps:**
 - Centrifugal vs. Reciprocating Pump. (W-20, W-23, W-24)
 - Working of Centrifugal Pump. (S-20, W-24, S-25)
 - Single Acting Reciprocating Pump working. (S-25)
3. **Priming:** What is Priming? Why is it required in Centrifugal pumps? (W-21, W-22, S-25, W-24)

Module 6: Refrigeration & Air Conditioning

6.1 Theory Questions

1. **VCR Cycle:** Explain Vapour Compression Refrigeration with schematic and charts. (W-21, W-22, S-23, S-25, W-24) [Image of Vapour Compression Refrigeration Cycle]
2. **VAR System:** Explain Vapour Absorption Refrigeration system. (W-23, S-24, S-25)

3. Air Conditioning:

- Working of Split AC. (S-23, W-20, S-25, W-24)
- Window Room AC. (W-23)
- Classification of Air Conditioning systems. (W-24)

4. Definitions: Refrigeration Effect, COP, Ton of Refrigeration (TR). (W-24, S-25)

Module 7: Couplings, Clutches & Brakes

7.1 Theory Questions**1. Difference:** Clutch vs. Brake vs. Coupling. (W-23, S-24, S-25)**2. Couplings:**

- Rigid vs. Flexible. (S-25, S-24)
- Flange, Universal, Oldham's coupling. (W-23, S-24)

3. Clutches: Single Plate Clutch (S-25), Centrifugal Clutch (W-24).**4. Brakes:** Internal Expanding Shoe Brake. (W-22, W-23, W-24)

Module 8: Transmission of Motion & Power

8.1 Theory Questions**1. Comparison:** Compare Belt, Chain, and Gear Drive. (W-21, S-20, W-23, S-24, W-24, S-25)**2. Pulleys:** Fast and Loose Pulley working. (W-22, W-23)**3. Gears:** Applications of Helical, Worm, Rack & Pinion. (W-24, W-23)

Module 9: Engineering Materials

9.1 Theory Questions**1. Definitions:** Elasticity, Plasticity, Ductility, Hardness, Toughness, Brittleness, Fatigue. (Recurring: W-21 to S-25)**2. Classification:** Ferrous vs. Non-Ferrous vs. Non-Metallic Materials. (W-21, W-22, W-24)**3. Specific Materials:** Properties of Cast Iron (W-23), Non-metallic materials (W-24).

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